

PATIENT MANAGEMENT SYSTEM

DEPLOYMENT USING DOCKER & KUBERNETES

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PROJECT OVERVIEW

The Patient Management System is a backend application developed using FastAPI that provides a RESTful API for managing patient records efficiently. It allows users to add, view, update, and delete patient information digitally. The system is lightweight, high-performance, and containerized using Docker, then deployed on Kubernetes for scalability and reliability

OBJECTIVE

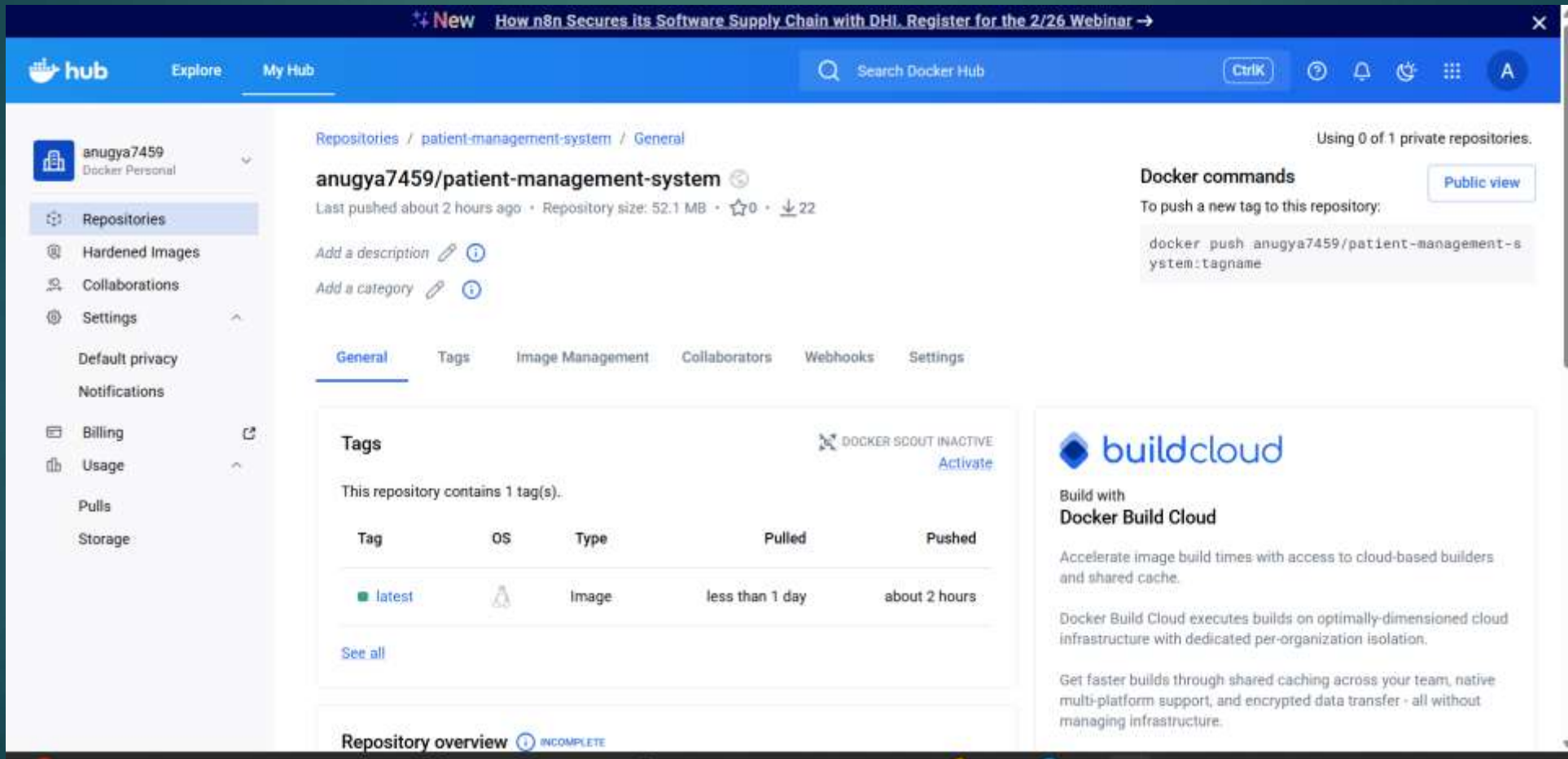
- ▶ **Efficient Data Management:** Digitally store and manage patient records to reduce manual work.
- ▶ **Fast Access:** Provide quick access to patient information for doctors and staff.
- ▶ **Data Accuracy:** Ensure accurate and organized patient data storage.
- ▶ **RESTful Backend:** Build a high-performance REST API using FastAPI for reliable data operations.
- ▶ **Scalability:** Deploy using Docker and Kubernetes to make the system scalable and portable.
- ▶ **Simplify Hospital Operations:** Make patient record handling easy, structured, and professional.

TECHNOLOGY USED

- **FastAPI (CRUD):** Used to create RESTful APIs for performing Create, Read, Update, and Delete operations on patient data.
- **Python:** Primary programming language used for backend logic and API development.
- **Docker:** Containerized the application for easy deployment and portability.
- **Kubernetes:** Orchestrated and managed containers to ensure scalability and reliability.
- **HTML, CSS, JavaScript:** Used to design and implement a user-friendly frontend interface for interacting with the system.

APPLICATION DEPLOYMENT & FRONTEND

- DOCKER HUB : Patient Management System image successfully pushed



The screenshot shows the Docker Hub interface for a repository named 'anugya7459/patient-management-system'. The repository is public and has been pushed about 2 hours ago. The repository size is 52.1 MB. The page displays the 'General' tab, which includes a table of tags. The 'latest' tag is listed with a pull time of 'less than 1 day' and a push time of 'about 2 hours'. The page also features a sidebar with navigation options, a search bar, and a 'Docker commands' section with a public view button. A 'buildcloud' advertisement is visible on the right side of the page.

Repositories / patient-management-system / General

anugya7459/patient-management-system

Last pushed about 2 hours ago • Repository size: 52.1 MB • ☆0 • ↓22

Add a description

Add a category

Using 0 of 1 private repositories.

Docker commands [Public view](#)

To push a new tag to this repository:

```
docker push anugya7459/patient-management-system:tagname
```

General Tags Image Management Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest	linux	Image	less than 1 day	about 2 hours

[See all](#)

Repository overview INCOMPLETE

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

KUBERNETES DASHBOARD

- Kubernetes: Application running with 1 Pod and NodePort service

The screenshot shows the Docker Desktop interface with the Kubernetes dashboard active. The top bar includes the Docker Desktop logo, a search bar, and system icons. The left sidebar contains navigation icons for various Docker features, with the Kubernetes icon highlighted. The main content area displays the following information:

Kubernetes [Give feedback](#)

Namespace: **default**

Cluster Stop Edit cluster

Cluster	Cluster type	Nodes	Kubernetes version
Active	kind	1	v1.31.1

Deployments

Name	Status	Pods
patient-app	Available	1/1

Pods

Name	Status
patient-app-7bc788464b-kpgqj	Running

At the bottom, a status bar indicates: **Kubernetes running** | RAM 2.64 GB | CPU 0.75% | Disk: 9.25 GB used (limit 1006.85 GB). On the right, there are links for **Terminal** and **Update available**.

Kubernetes [Give feedback](#)

Nodes

Name	Status
desktop-control-plane	Ready

Services

Name	Cluster IP	Ports
kubernetes	10.96.0.1	443/TCP
patient-service	10.96.249.135	80/TCP

Ingresses

No ingresses

Config maps and secrets

Name
kube-root-ca.crt

FRONTEND/WEBSITE

► Frontend: Patient Management System UI

 **Patient Management System**

[About](#) [Add Patient](#)

Filter by gender All Filter by age All Sort by Height ↑ Asc

Showing 7 patients 

ID	NAME	CITY	AGE	GENDER	HEIGHT (M)	WEIGHT (KG)	BMI	VERDICT	ACTIONS
P001	Ananya Verma	Guwahati	28	female	1.65	90	33.06	Obese	Edit Delete
P002	Ravi Mehta	Mumbai	35	male	1.75	85	27.76	Overweight	Edit Delete
P003	Sneha Kulkarni	Pune	22	female	1.6	45	17.58	Underweight	Edit Delete
P005	Aditya Jaiswal	Chauri Chaura	24	Other	0.75	90	160	Obesity	Edit Delete
P006	Priya Sharma	Delhi	30	female	1.58	55	22.03	Normal	Edit Delete

FEATURES OF THE SYSTEM

- **Add New Patients:** Easily add new patient details such as name, age, contact, and medical history.
- **View Patient Records:** Quickly view all patient records in an organized format.
- **Update Patient Information:** Modify existing patient data whenever required.
- **Delete Patient Data:** Remove unnecessary or outdated patient records safely.
- **Fast and Efficient:** REST API based backend ensures high performance and fast response.
- **User-friendly Interface:** Frontend built with HTML, CSS, and JavaScript for easy navigation.
- **Scalable & Portable:** Containerized with Docker and deployed on Kubernetes for scalability and reliability.

CONCLUSION

- The Patient Management System successfully digitizes patient records for efficient management.
- Built using FastAPI and REST APIs, the system is fast, lightweight, and reliable.
- Containerized with Docker and deployed on Kubernetes, ensuring scalability and portability.
- Provides a user-friendly interface for hospitals/clinics to manage patient information easily.
- Overall, the project demonstrates practical implementation of modern backend and deployment technologies.

PROJECT LINK

- ▶ GIT HUB LINK : <https://github.com/Anugya345/patient-management-system.git>
- ▶ DOCKER HUB (IMAGE) : [anugya7459/patient-management-system](https://hub.docker.com/r/anugya7459/patient-management-system)
- ▶ TO PULL : [docker pull anugya7459/patient-management-system:latest](#)
- ▶ [docker run -p 8000:8000 anugya7459/patient-management-system:latest](#)